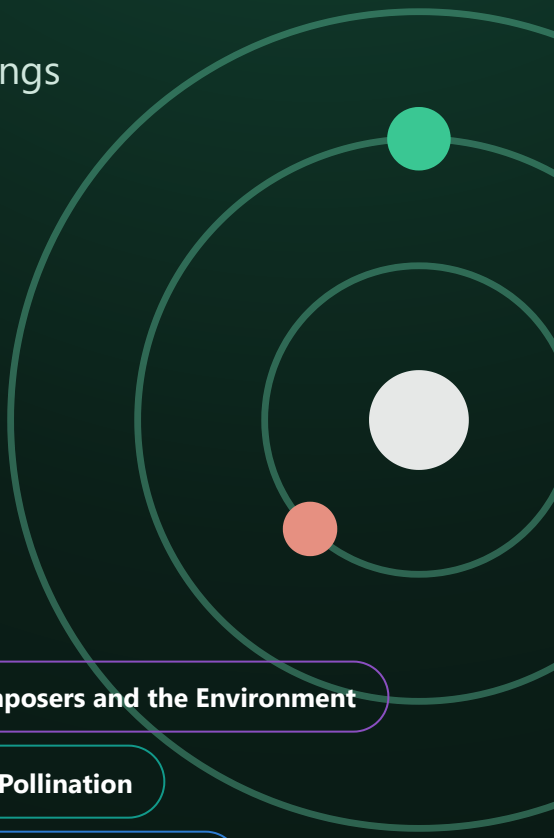
Discovery Lab

Exam 1 Revision

Lower Secondary Stage 1 · Biology · Lessons 1 to 9

Micro-organisms, plant life cycles, and the heart, lungs and circulation.



Lesson 1: Meet the Microbes

Lesson 2: Useful Microbes

Lesson 3: Harmful Microbes and Food Hygiene

Lesson 4: Decomposers and the Environment

Lesson 5: Germination and the Life Cycle

Lesson 6: Flowers and Pollination

Lesson 7: Fertilisation and Seed Dispersal

Lesson 8: The Heart, Blood and Vessels

Lesson 9: Lungs, Pulse and Exercise

Name: _____ Class: _____ Date: _____

LESSON 1

1 Meet the Microbes

Tiny living things that live everywhere.

Key ideas

Micro-organisms are tiny organisms usually seen with a **microscope**. Not all of them cause disease.

Bacteria are single-celled **prokaryotes**: their cells have **no nucleus**.

Viruses are much smaller than bacteria and can only reproduce inside a living **host cell**.

Microscopic fungi include **yeast** and **moulds**.

Antibiotic resistance: a few bacteria survive, multiply and pass on their protection.

Take antibiotics only when prescribed, finish the course and never share them.

Type	Cells?	Examples
Bacteria	One prokaryotic cell	Salmonella, Rhizobium
Viruses	No cells; need a host	Influenza, chickenpox
Fungi	Cells with a nucleus	Yeast, Penicillium

DID YOU KNOW?

A teaspoon of soil holds more bacteria than there are people on Earth.

Check yourself

- [ ] Name the three main types of micro-organism.
- [ ] Why are bacteria called prokaryotes?
- [ ] Why do viruses need a host cell?
- [ ] How does antibiotic resistance develop?
- [ ] Name two ways to prevent resistance.

REVISION TASK

Draw a bacterium, a virus and a yeast cell and label how they differ.

LESSON 2

2 Useful Microbes

Yeast, bacteria and fermentation.

Key ideas

Yeast feeds on sugar and makes **carbon dioxide**, so bread dough rises.

Bacteria turn milk into **yogurt** and some **cheeses**.

Fermentation is microbes breaking down sugars: kimchi, sauerkraut, soy sauce.

Penicillium mould led to **penicillin**, the first antibiotic.

Helpful **gut bacteria** help digest food.

Yeast balloon: yeast + sugar + warm water makes gas that inflates the balloon.

DID YOU KNOW?

Yeast has been used to make bread and beer for over 5,000 years.

Check yourself

- [ ] Which gas does yeast make?
- [ ] What is fermentation?
- [ ] Which microbes make yogurt?
- [ ] Which mould led to penicillin?
- [ ] Why does too much heat stop the yeast balloon?

REVISION TASK

Draw the yeast balloon experiment and label the yeast, sugar, warm water and gas.

LESSON 3

3 Harmful Microbes and Food Hygiene

Disease, food poisoning and how to stop microbes spreading.

Key ideas

Salmonella causes food poisoning; **Staphylococcus** causes skin infections.

Viruses cause **influenza**, **chickenpox** and **COVID-19**.

Cross-contamination: microbes pass from one food or surface to another.

Wash hands with soap for at least **20 seconds**.

Cook food thoroughly and keep raw and cooked food **separate**.

A fridge **slows** bacteria; it does **not** kill them all.

Disease	Cause
Food poisoning	Salmonella (bacteria)
Strep throat	Bacteria
Influenza	Virus
Chickenpox	Virus

DID YOU KNOW?

Over 600 million people get sick from contaminated food each year.

Check yourself

- [ ] Which bacterium causes food poisoning?
- [ ] What is cross-contamination?
- [ ] How long should you wash hands?
- [ ] Does refrigeration kill all bacteria?
- [ ] Name three ways to make food preparation safer.

REVISION TASK

Make a poster with five rules for safe food preparation.

LESSON 4

4 Decomposers and the Environment

Nature's recyclers, and microbes in soil and water.

Key ideas

Decomposers (bacteria and fungi) break down dead plants and animals.

They **recycle nutrients** into the soil for plants to absorb.

Composting uses decomposers to turn food scraps into compost.

Rhizobium in legume roots **fixes nitrogen** for the plant.

Bacteria in **water treatment** plants break down waste.

Some microbes live in **extreme** places such as hot springs and the deep sea.

DID YOU KNOW?

One gram of healthy soil can hold up to a billion bacteria.

Check yourself

- [ ] What is a decomposer?
- [ ] What happens to nutrients when a leaf decomposes?
- [ ] What does Rhizobium do?
- [ ] How do bacteria help clean water?
- [ ] What would happen without decomposers?

REVISION TASK

Draw the nutrient cycle: dead leaf, decomposers, soil, plant roots.

LESSON 5

5 Germination and the Life Cycle

How a seed wakes up and starts a new plant.

Key ideas

Life cycle: **germination, flowering, pollination, fertilisation, seed dispersal**.

Germination is a seed starting to grow into a new plant.

Seeds need **water**, **warmth** and **oxygen**, but not direct sunlight.

Water softens the seed coat; warmth speeds up reactions; oxygen is for **respiration**.

Dormancy: a seed stays inactive until conditions are right.

A fair test changes **one** variable and keeps the others the same.

DID YOU KNOW?

Some desert seeds only germinate after heavy rain, which may happen once a year.

Check yourself

- [ ] What is germination?
- [ ] Which three conditions do seeds need?
- [ ] Do seeds need sunlight to germinate?
- [ ] What is dormancy?
- [ ] How would you test the effect of water?

REVISION TASK

Plan an experiment on water and germination: variable, measurement, controls.

LESSON 6

6 Flowers and Pollination

Flower parts, and how pollen gets from flower to flower.

Key ideas

Anther: makes pollen. **Stigma**: sticky, catches pollen. **Ovary**: becomes the fruit. **Petals**: attract pollinators.

Pollination is the transfer of pollen from an anther to a stigma.

Insect flowers: bright petals, sweet smell, nectar.

Wind flowers: small dull petals, lots of light pollen, feathery stigmas.

Bees collect nectar; pollen sticks to them and rubs off on the next stigma.

Examples: sunflower (insect), wheat (wind).

Feature	Insect	Wind
Petals	Large, bright	Small, dull
Pollen	Sticky	Light, lots
Stigma	Sticky, inside	Feathery, outside

DID YOU KNOW?

The Rafflesia arnoldii is the biggest flower in the world: up to 1 metre wide.

Check yourself

- [ ] Which part makes pollen?
- [ ] Which part catches pollen?
- [ ] Name two features of an insect-pollinated flower.
- [ ] Why do wind-pollinated flowers make so much pollen?
- [ ] How does a bee help a plant reproduce?

REVISION TASK

Draw a flower and label the anther, stigma, ovary and petals.

LESSON 7

7 Fertilisation and Seed Dispersal

From pollen to seeds, then off to new places.

Key ideas

After pollination, male and female cells **unite**: **fertilisation** forms a **seed**.

The **ovary** usually becomes the **fruit**, which protects the seeds.

Dispersal methods: wind, water, animals and exploding pods.

Dandelion seeds have **parachutes**; coconuts **float**; berries are **eaten**; some pods **burst**.

Maple seeds **spin** to stay in the air longer.

Dispersal reduces **competition** and lets plants **colonise** new areas.

DID YOU KNOW?

Bananas can reproduce without seeds: this is called vegetative reproduction.

Check yourself

- [ ] What is fertilisation?
- [ ] What does the ovary become?
- [ ] Name four ways seeds are dispersed.
- [ ] Why do maple seeds spin?
- [ ] Why is dispersal important?

REVISION TASK

Draw a seed and its dispersal method for each of wind, water, animals and explosion.

LESSON 8

8 The Heart, Blood and Vessels

The body's transport system.

Key ideas

The circulatory system moves **oxygen, nutrients and waste**. Parts: heart, arteries, veins, blood (and capillaries).

The heart has **four chambers**: two atria and two ventricles.

Left side: oxygenated blood to the body. **Right** side: deoxygenated blood to the lungs.

Blood: **red cells** carry oxygen; **white cells** fight infection; **platelets** clot; **plasma** is the liquid.

Arteries carry blood away from the heart; **veins** carry it back and have valves; **capillaries** are one cell thick.

Pulmonary: heart to lungs. **Systemic**: heart to body. **Coronary**: feeds the heart muscle.

Vessel	Direction	Feature
Artery	Away from the heart	Thick strong wall
Vein	To the heart	Valves
Capillary	Between them	One cell thick

DID YOU KNOW?

Your heart beats about 100,000 times every day.

Check yourself

- [ ] How many chambers does the heart have?
- [ ] What do red blood cells do?
- [ ] Which vessels have valves?
- [ ] Why are capillary walls one cell thick?
- [ ] What is coronary circulation?

REVISION TASK

Draw the heart with four chambers and add arrows for blood to the lungs and to the body.

LESSON 9

9 Lungs, Pulse and Exercise

Gas exchange, breathing, respiration and the pulse.

Key ideas

Alveoli are tiny air sacs where **oxygen enters** the blood and **carbon dioxide leaves**.

Breathing moves air (physical). **Respiration** releases energy in cells (chemical, in mitochondria).

Pulse is the beat of the arteries as the heart pumps.

Pulse per minute = beats in **15 seconds × 4**.

Exercise raises **pulse and breathing rates** so muscles get more oxygen.

Athletes often have a **lower resting** pulse.

DID YOU KNOW?

Blood takes less than a minute to travel all the way round your body.

Check yourself

- [ ] What happens in the alveoli?
- [ ] What is the difference between breathing and respiration?
- [ ] 24 beats in 15 seconds is how many per minute?
- [ ] Why does breathing rate rise during exercise?
- [ ] Why does the pulse slowly return to normal after exercise?

REVISION TASK

Design a pulse experiment: variable, controls, how you will record results and make them reliable.

FINAL PRACTICE

Final practice

Tick each box when you can answer without looking.

Final practice

- [ ] Explain why viruses need living host cells to reproduce, and how antibiotic resistance can develop.
- [ ] A student makes a sandwich after touching raw meat. Explain the risks and three safer actions.
- [ ] Compare insect pollination and wind pollination: explain two differences in how flowers are adapted.
- [ ] Explain why seed dispersal matters, including competition and new areas.
- [ ] Compare arteries, veins and capillaries and explain how each is adapted.

CHALLENGE

Link two topics: explain how oxygen travels from the air to your muscle cells, and why your pulse and breathing rise when you run.

Revision checklist

Revision task	Done?
Reviewed the Canvas slides for the three units	☐
Answered every "Check yourself" question	☐
Practised the matching and sorting activities	☐
Used the 3D labs: yeast, germination, pollination and heart	☐
Practised long answers and the practice exam	☐
Reviewed keywords: fermentation, germination, pollination, fertilisation, pulse	☐

Try it live · Practise with 3D labs, games and a practice exam at **morehumaned.com/discovery-lab**

ANSWER KEY

Answer key

Check your own answers against these. If one is hard, go back to the lesson.

Lesson 1: Meet the Microbes

1. Bacteria, viruses and microscopic fungi.
2. Their cells have no nucleus.
3. They can only copy themselves inside a living cell.
4. Survivors with a chance change multiply and pass it on.
5. Finish the course; never share antibiotics (or take them only when prescribed).

Lesson 2: Useful Microbes

1. Carbon dioxide.
2. Microbes breaking down sugars.
3. Bacteria.
4. Penicillium.
5. High heat kills the yeast.

Lesson 3: Harmful Microbes and Food Hygiene

1. Salmonella.
2. Harmful microbes passing between foods or surfaces.
3. At least 20 seconds with soap.
4. No: it only slows their growth.
5. Wash hands, separate raw and cooked foods, cook thoroughly (also: fridge, clean boards).

Lesson 4: Decomposers and the Environment

1. An organism (bacteria or fungus) that breaks down dead matter.
2. They return to the soil for plants to use.
3. It fixes nitrogen for legume plants.
4. They break down waste in the water.
5. Dead matter would pile up and nutrients would run out.

Lesson 5: Germination and the Life Cycle

1. A seed starting to grow into a plant.
2. Water, warmth and oxygen.
3. No.
4. A seed staying inactive until conditions suit it.
5. Give groups different amounts of water, keep everything else the same and count germinated seeds.

ANSWER KEY

Answer key

Check your own answers against these. If one is hard, go back to the lesson.

Lesson 6: Flowers and Pollination

1. The anther.
2. The stigma.
3. Bright petals and nectar (or sweet smell).
4. Most is wasted, so some must reach a stigma.
5. It carries pollen on its body from flower to flower.

Lesson 7: Fertilisation and Seed Dispersal

1. Male and female cells joining to form a seed.
2. A fruit.
3. Wind, water, animals, explosion.
4. To stay in the air longer and travel farther.
5. It reduces competition and helps plants reach new areas.

Lesson 8: The Heart, Blood and Vessels

1. Four.
2. Carry oxygen.
3. Veins.
4. To allow quick exchange with tissues.
5. Blood supply to the heart muscle itself.

Lesson 9: Lungs, Pulse and Exercise

1. Gas exchange: oxygen into the blood, carbon dioxide out.
2. Breathing moves air; respiration releases energy in cells.
3. 96 beats per minute.
4. To take in more oxygen and remove more carbon dioxide.
5. Muscles need less oxygen once you rest.

Final practice

1. Viruses need a host to copy themselves; survivors of an antibiotic multiply and pass on resistance.
2. Bacteria spread from raw meat to the sandwich (cross-contamination); wash hands, separate boards, cook meat thoroughly.
3. Insect flowers: bright petals, nectar, sticky pollen; wind flowers: small petals, lots of light pollen, feathery stigmas.
4. It reduces competition for light, water and nutrients and lets plants colonise new places.
5. Arteries: thick walls, away from the heart; veins: valves, towards the heart; capillaries: one cell thick for exchange.

Discovery Lab

You are going to do great! Use this booklet to plan your study time, then bring it all together.

Mr Guevara · morehumaned.com/discovery-lab